

1. Connector & input protection

Field connector, 2-stage bidirectional
TVS surge protection, chassis
isolation

: HMT500(260313A)_connector.kicad_sch

2. Power

LDO 3.3V

: HMT500(260313A)_power.kicad_sch

3. MCU

STM32G0B1CCT6, SWD, reset, internal
temperature sensor

: HMT500(260313A)_mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04)
and Pt1000 2-wire (ADS1220)

: HMT500(260313A)_measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611
miswiring protection

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: HMT500(260313A)_analog_out.kicad_sch
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6. RS-485

THVD2410 $\pm 70\text{V}$ fault-protected
transceiver

: HMT500(260313A)_rs485.kicad_sch

7. Aux

AO supply VAO 16.1 V (buck), supply
monitor

: HMT500(260313A)_aux.kicad_sch

DOTECH HMT500(260313) - Oil Moisture Transmitter - Schematic v0.9 - Project No. HMT500(260313A)

File names (schematic, PCB, Gerber) = HMT500(260313A).* PCB silkscreen marking = HMT500(260313A)

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2410 -> J1

Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-28 V DC.

v0.10: AO D30/D40 TVS1401 ± 14 V. AO $\pm 28/30$ V continuous miswiring is NOT guaranteed (#41).

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels.

History v0.3-v0.8 (MCU G0B1, JST harness, DAC8760/TPS26611 datasheet pinouts, RTN isolation, gated DAC SCLK ...): docs/decision-log.md #9-#26.

v0.9: 3-pass review (docs/hw/final-review-260313A.md): OVP 32.6 V (R5 36.5k); AO supply VAO 16.1 V buck U15 (DAC AVDD, IPS26611 +Vs; dec. #37, IVS320 AO); D3 series Schottky; RS-485 TVS D60/D61; D2 -> TVS3301; eFuse IN C2; buck CIN;

RS485_DE 10k pull-down (in-house bootloader, dec. #31); THVD2410; X7R only; R2 HV 1206; SGOOD -> PA0/PA4; CS_CDC -> PA15; VIN_SENSE PB0;

LATCH/CS pull-ups; OPA197 V+ = DAC AVDD; R31/R41 100k; SWD no-connect flags removed; sheet Aux: +Vs clamp, VIN monitor (DAC boost / HSE / sensor ESD: no room, dec. #32).

Footprints: KiCad 7.0.11 library + project library HMT500_260313A (GDT Bourns 2035-xx-SM, TVS3301 pad floating).

표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울인 글자 (+3V3, VIN_P)

신호 네트 = 테두리 있는 라벨 (SPI_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음

라벨 모양 = 신호 방향: 뾰족한 쪽이 밖 = 이 시트에서 내보냄(출력), 안 = 받음(입력), 양쪽 = 양방향, 네모 = 아날로그·수동

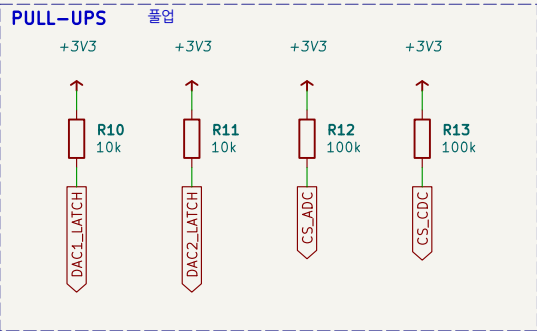
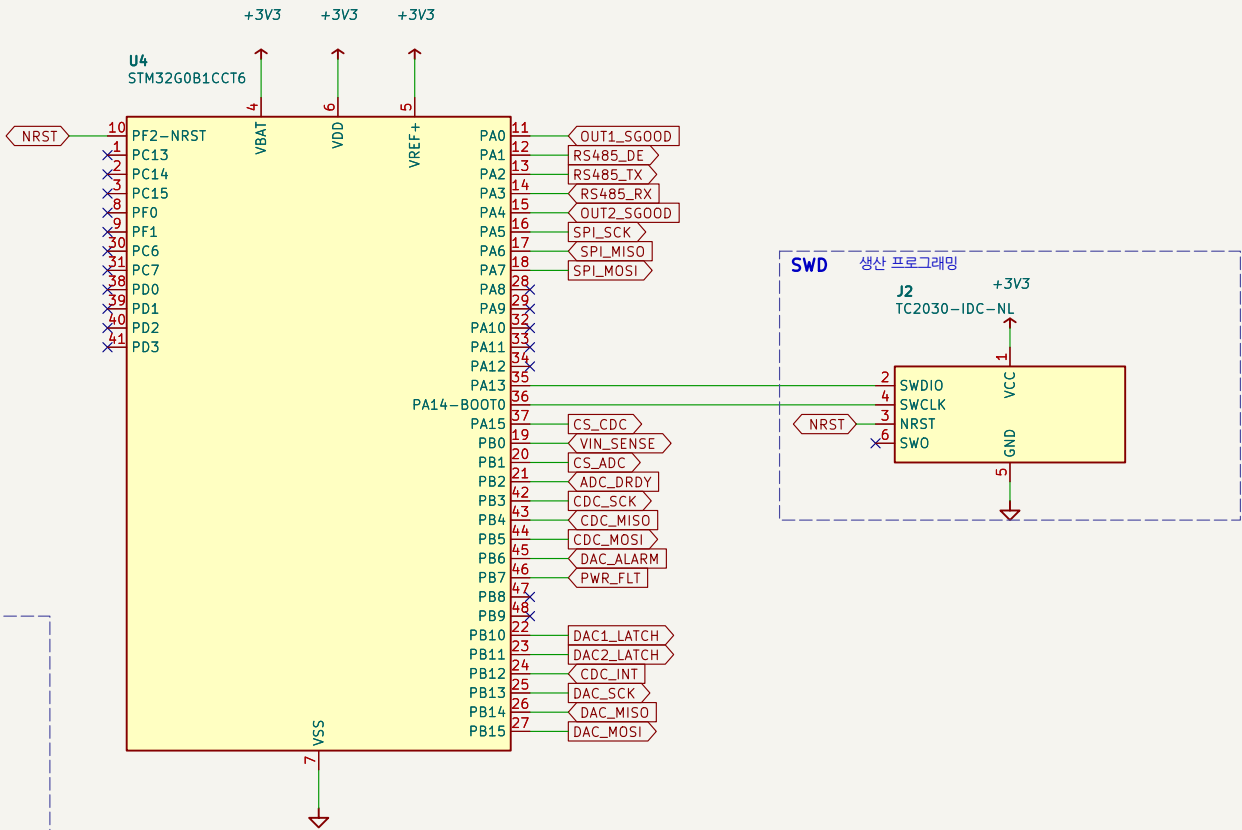
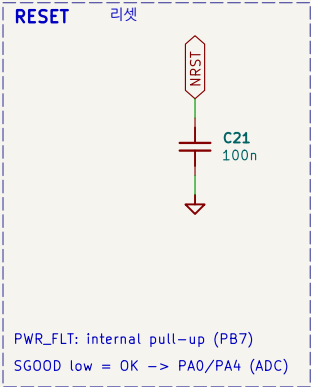
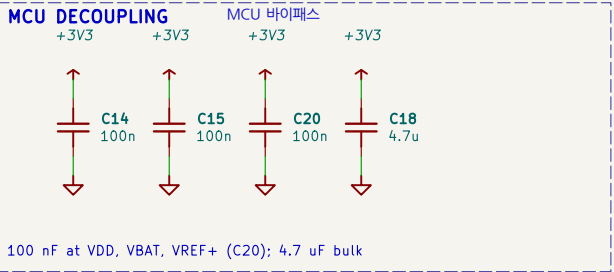
글자 겹침: 생성기가 부품 몸체·핀·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
DOTEC Co., Ltd.

Sheet: /
File: HMT500(260313A).kicad_sch

Title: HMT500(260313A) Top

Size: A3	Date: 2026-10-01	Rev: 0.10
KiCad E.D.A. 10.0.6		Id: 1/8



MCU NOTES (STM32G0B1) MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1, IWDG_SW=0, RDP 1

-> in-house RS-485 bootloader 0x08000000 (dec. #31).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

VIN_SENSE (PB0 ADC_IN8) = VIN_P / 11.

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
U4	STM32G0B1CCT6	계어통신·보정 연산	Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40-85 °C (샘플 #39), LQFP48, 내부 온도센서 (PCB 온도), SPI1 = ADS1220, SPI2 = DAC (모드 3, SCLK 게이트), SPI3 = PCAP04
J2	TC2030-IDC-NL	생산 때 펌웨어 기록 (SWD)	PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더
C21	100n	리셋 노이즈 필터	NRST 핀, 물딩 후 EMC 여유
R10, R11	10k	DAC LATCH 풀업	리셋 중 LATCH High -> SCLK 게이트 닫힘, DAC 오입력 방지 (v0.9)
R12, R13	100k	CS 풀업	리셋 중 ADS1220-PCAP04 선택 해제 (디지털 입력 부동 금지)

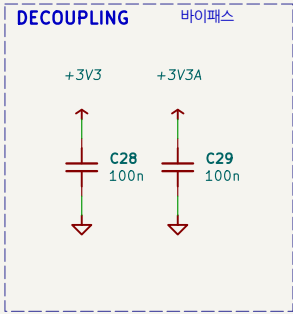
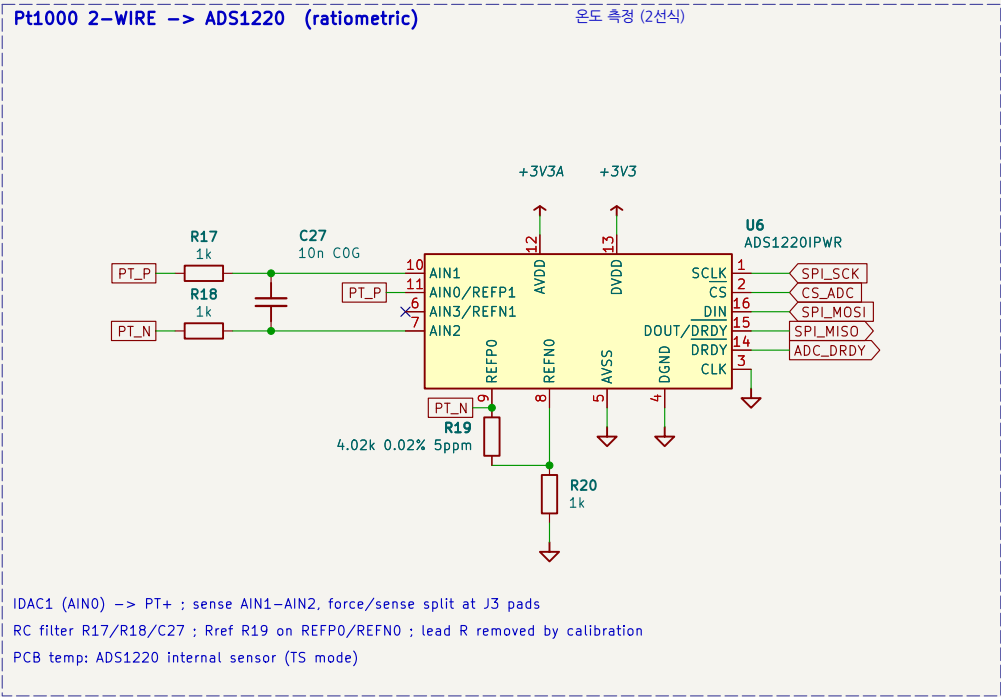
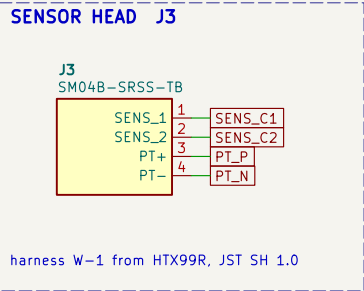
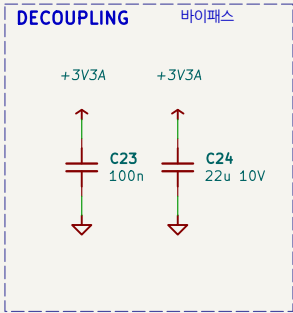
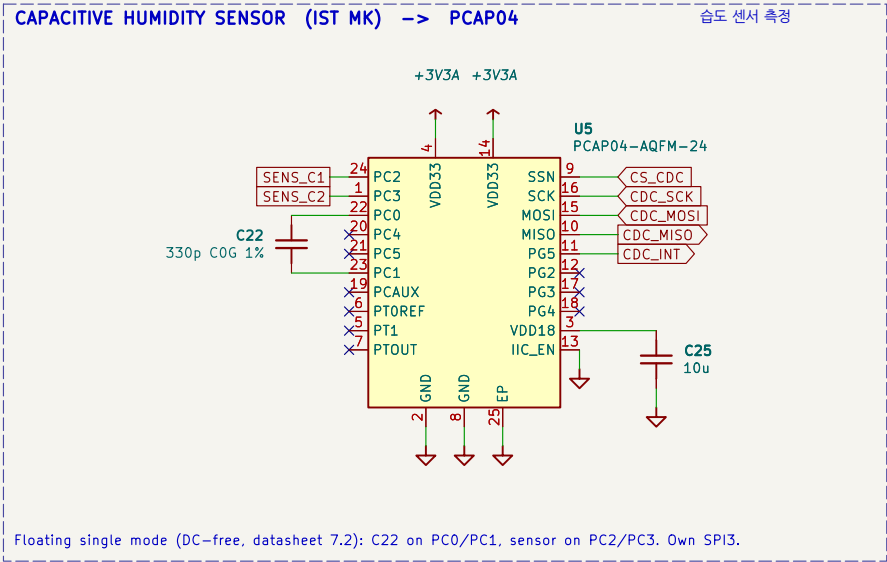
* 표시 값은 데이터시트로 확인

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Sheet: /3. MCU/
File: HMT500(260313A)_mcu.kicad_sch

Title: HMT500(260313A) MCU

Size: A3	Date: 2026-10-01	Rev: 0.10
KiCad E.D.A. 10.0.6		Id: 4/8



주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
J3	SM04B-SRSS-TB	센서 하네스 W-1 연결	JST SH 1.0 mm 4P 옆 삽입, 1:2 = MK33, 3:4 = Pt1000
U5	PCAP04-AQFM-24	습도 센서 정전용량 측정	정전용량-디지털 변환, 기준 C와 비율 측정(방전 시간), DC 없는 구동, 내부 DSP, 전용 SPI3 (모드 1)
C25, C24	10u / 22u	PCAP04 전원 버퍼	VDD18 ≥ 4.7 uF, VDD33 ≥ 10 uF (데이터시트 표 2) — DC 바이어스 후에도 총축 (v0.9)
C22	330p COG 1%	PCAP04 기준 용량	센서 MK33-W (300 pF, 결정 #33) 측정 범위 가운데 값, COG
U6	ADS1220IPWR	Pt1000 온도 측정	24비트 ADC, 50/60 Hz 제거, 내부 온도센서 (PCB 온도), 설정: IDAC = 250 uA 고정 (100 uA면 기준 0.40 V < 권장 0.75 V), PGA ≤ 2 (SP1 모드 1)
R19	4.02k 0.02% 5ppm	Pt1000 비율 측정 기준저항	IDAC 전류 오차 상쇄, 온도계수 5 ppm/°C
R17, R18, C27	1k / 10n COG	입력 RC 필터	차동 노이즈 제거 (COG, 데이터시트 9.2)
* 표시 같은 데이터시트로 확인			

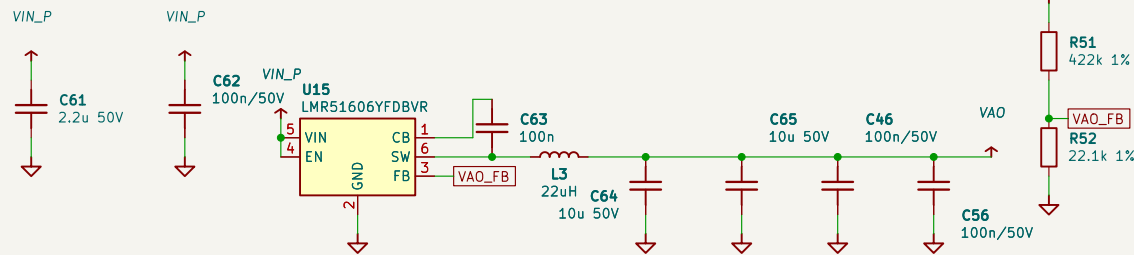
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DOTECH Co., Ltd.

Sheet: /4. Measurement/
File: HMT500(260313A)_measurement.kicad_sch

Title: HMT500(260313A) Measurement

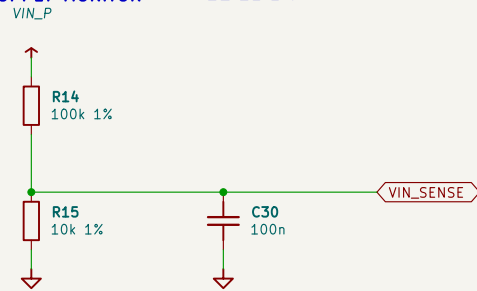
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AO SUPPLY VAO 16.1 V - DC-DC (decision #37)



SUPPLY MONITOR

전원 전압 감시



VIN_SENSE = VIN_P / 11 -> PB0 (ADC_IN8)

OPTIONS NOT FITTED ON PCB

PCB에 넣지 않은 선택 사항

DAC8760 IOUT boost: not needed – AVDD = VAO 16.1 V (decision #37) cuts DAC heat

from about 0.66 W to 0.39 W per channel (24 mA short, any supply 12–28 V).

Thermal test on a potted prototype still planned.

HSE crystal: no room next to PF0/PF1 -> factory HSI trim + firmware HSI calibration

from Modbus frame timing (USART auto-baud / timer capture).

Sensor-line ESD array: no room next to J3 -> probe ESD test (IEC 61000-4-2) decides;

if needed: protect at the HTX99R side or in the next PCB rev.

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
U15	LMR51606YFDBVR	AO 전원 VAO 16.1 V 벅	4-65 V 입력 0.6 A, 1.1 MHz FPWM, SOT-23-6 — DAC AVDD-TPS26611 +Vs-OPA197 V+ 공용 (IVS320 AO rev 1.0과 같은 회로, 결정 #37)
L3, C64, C65	22uH / 10u 50V x2	벅 출력	22 uH 1.05 A 4 × 4 × 2 mm, 10 uF 50 V X7R 1206 × 2 (16 V에서 실효 약 4 uF × 2)
R51, R52	422k / 22.1k 1%	VAO 설정	$0.8 \text{ V} \times (1 + 422/22.1) = 16.08 \text{ V}$
C61, C62, C63	2.2u 50V / 100n / 100n	입력-부트스트랩	VIN 핀 옆 입력 C, CB-SW 100 nF
C46, C56	100n/50V	+Vs 바이패스	C46 → U9 6번, C56 → U10 6번 핀 옆
R14, R15, C30	100k / 10k / 100n	전원 전압 감시	VIN_P / 11 → PB0 ADC (UVLO 8.9 V ~ DAC 최소 10 V 사이 판정)

* 표시 값은 데이터시트로 확인

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DOTECH Co., Ltd.

Sheet: /7. Aux/
File: HMT500(260313A)_aux.kicad_sch

Title: HMT500(260313A) Aux

Size: A3	Date: 2026-10-01	Rev: 0.10
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